MERCK MANUAL
Professional Version

Hypoglycemia

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Last updated: Dec 2025

Hypoglycemia, or low plasma glucose level can result in sympathetic nervous system stimulation and central nervous system dysfunction. In patients with diabetes who take insulin or antihyperglycemic treatment, hypoglycemia is common and is defined as a glucose level of ≤ 70 mg/dL. In contrast, hypoglycemia unrelated to exogenous insulin therapy is an uncommon clinical syndrome caused by various disorders or medications. Diagnosis requires blood tests performed at the time of symptoms or during a 72-hour fast. Treatment of hypoglycemia is provision of glucose combined with treatment of the underlying disorder.

Most commonly, symptomatic hypoglycemia is a complication of medication treatment of diabetes mellitus with oral antihyperglycemics (especially sulfonylureas) and insulin.

Symptomatic hypoglycemia unrelated to treatment of diabetes mellitus is relatively rare, in part because the body has extensive counter-regulatory mechanisms to compensate for low blood glucose levels. Glucagon and epinephrine levels surge in response to acute hypoglycemia and appear to be the first line of defense. Cortisol and growth hormone levels also increase acutely and are important in the recovery from prolonged hypoglycemia. The threshold for release of these hormones is usually above that for hypoglycemic symptoms. Hereditary or congenital syndromes that cause hypoglycemia in infancy and childhood are not discussed here. (See Neonatal Hypoglycemia and Overview of Carbohydrate Metabolism Disorders.)

Etiology of Hypoglycemia

Hypoglycemia in patients without diabetes is rare (1). It can occur in the fasting or in the postprandial state (1 to 3 hours after eating) and can be classified as insulin-mediated or non-insulin–mediated. A helpful practical classification is based on clinical status: whether hypoglycemia occurs in patients who appear healthy or ill.

Well-appearing patients

In well-appearing adults without diabetes, the differential diagnosis includes insulin-mediated and non-insulin–mediated disorders (2).

Insulin-mediated causes include:

- Exogenous insulin
- Insulin secretagogue (sulfonylurea) use
- Insulinoma
- Post-bariatric surgery hypoglycemia
- Non-insulinoma pancreatogenous hypoglycemia syndrome (NIPHS)
- Insulin autoimmune hypoglycemia

Insulinoma is a rare neuroendocrine tumor of insulin-producing beta cells. It typically causes fasting hypoglycemia although postprandial hypoglycemia can also occur.

Hypoglycemia that occurs after bariatric surgery is a hyperinsulinemic hypoglycemia that develops sometimes years after bariatric (especially roux-en-Y gastric bypass) surgery. It is rare, but the true frequency is not known, and pathology specimens show nesidioblastosis (abnormal proliferation of pancreatic beta cells). Hypoglycemia is typically postprandial.

NIPHS is a rare condition in which patients have hyperinsulinemic hypoglycemia with negative localizing imaging studies and no history of bariatric surgery. Hypoglycemia is usually postprandial, and surgical specimens demonstrate features of nesidioblastosis.

Insulin autoimmune hypoglycemia is a condition that most often occurs in patients with other autoimmune conditions such as systemic lupus erythematosus. Autoantibodies bind to insulin or insulin receptors on cell membranes in target tissues (eg, the liver, muscles, adipose tissue) and dissociate. Circulating insulin becomes available to bind to the receptor upon dissociation from the antibodies, causing hypoglycemia. Treatment is corticosteroids or immunosuppressants.

Non-insulin–mediated causes include:

- Adrenal insufficiency
- Use of medications other than insulin or a sulfonylurea (eg, quinine, gatifloxacin, pentamidine, alcohol)
- Congenital causes of hypoglycemia (eg, glycogen storage disorders)

Ill-appearing patients

In ill patients, the differential diagnosis also includes insulin-mediated and non-insulin–mediated disorders (2).

Insulin-mediated disorders include:

- Exogenous insulin
- Insulin secretagogue (sulfonylurea) use

Non-insulin–mediated disorders include:

- Undernutrition or starvation
- Cirrhosis
- Sepsis
- End-stage renal disease
- Heart failure, if advanced
- Adrenal insufficiency
- Non-islet cell tumor hypoglycemia
- Use of medications other than insulin or a sulfonylurea

In patients who are ill and hospitalized, spontaneous hypoglycemia that is not caused by medications can occur when poor nutrition is combined with advanced organ failure (especially liver, kidney, or heart failure) and/or sepsis. In these patients, hypoglycemia portends a poor prognosis.

Non-islet cell tumor hypoglycemia is a rare condition caused by production of large amounts of aberrant forms of insulin-like growth factor 2 (IGF-2) by a tumor. The aberrant form of IGF-2 binds to the insulin receptor and causes hypoglycemia. By the time hypoglycemia develops, the tumor is usually advanced.

Factitious hypoglycemia is true hypoglycemia induced by nontherapeutic administration of sulfonylureas or insulin.

Pseudohypoglycemia

Pseudohypoglycemia occurs when processing of blood specimens in untreated test tubes is delayed and cells, such as red blood cells and leukocytes (especially if increased, as in leukemia or polycythemia), consume glucose. Poor circulation to the digits can also cause erroneously low fingerstick glucose measurements.

Etiology references

1. González-Vidal T, Lado-Baleato Ó, Fernández-Merino C, et al. Evaluation of Asymptomatic Fasting Hypoglycemia in Outpatients Without Diabetes. *J Am Board Fam Med.* Published online August 11, 2025. doi:10.3122/jabfm.2024.240274R1
2. Looi E, Lawler HM. Non-Diabetic Hypoglycemia: Evaluation and Management in Adults. *J Clin Med.* 2025;14(13):4393. doi:10.3390/jcm14134393

Symptoms and Signs of Hypoglycemia

The surge in autonomic activity in response to low plasma glucose causes sweating, nausea, warmth, anxiety, tremulousness, palpitations, and possibly hunger and paresthesias (1). Insufficient glucose supply to the brain (neuroglycopenia) causes headache, blurred or double vision, confusion, agitation, seizures, and coma. In older patients, hypoglycemia may cause stroke-like symptoms of aphasia or hemiparesis and is more likely to precipitate stroke, myocardial infarction, and sudden death (2, 3).

Patients with diabetes mellitus, especially patients with type 1 diabetes, type 2 diabetes of long duration or patients with frequent hypoglycemia, may be unaware of hypoglycemic episodes because they no longer experience autonomic symptoms (hypoglycemia unawareness) (4).

In research studies under controlled conditions, autonomic symptoms begin at or beneath a plasma glucose level of approximately 60 mg/dL (3.3 mmol/L), whereas central nervous system symptoms occur at or below a glucose level of approximately 50 mg/dL (2.8 mmol/L) (5). People with glucose levels at these thresholds may have no symptoms, while people with symptoms suggestive of hypoglycemia can have normal glucose concentrations.

Symptoms and signs references

1. Palani G, Stortz E, Moheet A. Clinical Presentation and Diagnostic Approach to Hypoglycemia in Adults Without Diabetes Mellitus. *Endocr Pract.* 2023;29(4):286-294. doi:10.1016/j.eprac.2022.11.010
2. Makam S, Stein LK, Dhamoon MS. Hypoglycemic Events May Trigger Acute Ischemic Stroke Within 30 Days in Those With Diabetes: A Case-Crossover Study. *Stroke.* 2025;56(1):122-127. doi:10.1161/STROKEAHA.124.049178
3. Mattishent K, Loke YK. Meta-Analysis: Association Between Hypoglycemia and Serious Adverse Events in Older Patients Treated With Glucose-Lowering Agents. *Front Endocrinol (Lausanne).* 2021;12:571568. doi:10.3389/fendo.2021.571568
4. Martín-Timón I, Del Cañizo-Gómez FJ. Mechanisms of hypoglycemia unawareness and implications in diabetic patients. *World J Diabetes.* 2015;6(7):912-926. doi:10.4239/wjd.v6.i7.912
5. Cryer PE. Symptoms of hypoglycemia, thresholds for their occurrence, and hypoglycemia unawareness. *Endocrinol Metab Clin North Am.* 1999;28(3):495-vi. doi:10.1016/s0889-8529(05)70084-0

Diagnosis of Hypoglycemia

- Measurement of blood glucose level
- Sometimes fasting glucose
- Sometimes serum insulin, insulin antibodies, sulfonylurea levels, C-peptide, proinsulin, serum ketones

In patients with diabetes who are taking insulin or antihyperglycemic medications, a blood glucose level < 70 mg/dL (3.9 mmol/L) correlated with clinical findings is consistent with hypoglycemia.

The severity of hypoglycemia in patients with diabetes is based on blood glucose levels and need for assistance:

- Level 1 (mild) hypoglycemia: Blood glucose < 70 mg/dL (< 3.9 mmol/L) but ≥ 54 mg/dL (≥ 3 mmol/L)

- Level 2 (moderate) hypoglycemia: Blood glucose < 54 mg/dL (< 3 mmol/L)
- Level 3 (severe) hypoglycemia: Hypoglycemia requiring assistance from another person due to change in mental or physical status

In patients who are not receiving diabetes treatment, diagnosis of a hypoglycemic disorder requires confirmation of **Whipple's triad** or confirmation of low blood glucose during a fast. Whipple's triad includes:

- Symptoms of hypoglycemia
- Low plasma glucose level (< 55 mg/dL [3.05 mmol/L]) that occurs at the time symptoms occur
- Decrease in symptoms when dextrose or another sugar is given

If a clinician is present when symptoms occur, blood should be sent for glucose testing in a tube containing a glycolytic inhibitor. If glucose is normal, hypoglycemia is excluded and other causes of the symptoms should be considered. If glucose is abnormally low and no cause can be identified from history (eg, medications, adrenal insufficiency, severe undernutrition, organ failure ,sepsis), serum insulin, insulin antibodies, and sulfonylurea levels should be checked. C-peptide and proinsulin measured from the same tube can distinguish insulin-mediated from non–insulin-mediated and factitious from physiologic hypoglycemia and can obviate the need for further testing.

In practice, however, it is unusual that clinicians are present when patients experience symptoms suggestive of hypoglycemia. Home glucose meters are unreliable for quantifying hypoglycemia, and there are no clear glycosylated hemoglobin (HbA1C) thresholds that distinguish long-term hypoglycemia from normoglycemia. Continuous glucose monitors are also less accurate in the hypoglycemic range. So the need for more extensive diagnostic testing is based on the probability that an underlying disorder that could cause hypoglycemia exists given a patient’s clinical appearance and coexisting illnesses.

To differentiate between insulin-mediated and non-insulin–mediated hypoglycemia and to determine the etiology of hypoglycemia, a 48- or 72-hour fast may be required. A prolonged fast can trigger hypoglycemia in a person who has a history of fasting hypoglycemia.

A 72-hour fast done in a controlled setting is the standard for diagnosis. However, in almost all patients with a hypoglycemic disorder, a 48-hour fast is adequate to detect hypoglycemia, and a full 72-hour fast may not be necessary. Patients drink only noncaloric, noncaffeinated beverages. Plasma glucose is measured at baseline, whenever symptoms occur, and every 4 to 6 hours or every 1 to 2 hours if glucose falls below 70 mg/dL (3.9 mmol/L).

Serum insulin, C-peptide, and proinsulin should be measured when a simultaneous plasma glucose measurement is < 55 mg/dL (< 3.05 mmol/L). These measurements help to distinguish endogenous from exogenous (factitious) hypoglycemia. The fast is terminated at 72 hours if the patient has experienced no symptoms and glucose remains normal, sooner if glucose decreases to ≤ 45 mg/dL (≤ 2.5 mmol/L) in the presence of symptoms of hypoglycemia.

End-of-fast measurements include beta-hydroxybutyrate (which should be low if the cause is an insulinoma), serum sulfonylurea to detect medication-induced hypoglycemia, and plasma glucose after

IV glucagon injection to detect an increase characteristic of insulinoma. Sensitivity, specificity, and predictive values for detecting hypoglycemia by this protocol have not been reported.

If symptomatic hypoglycemia has not occurred by 48 to 72 hours, the patient should exercise vigorously for about 30 minutes. If hypoglycemia still does not occur, insulinoma is essentially excluded and further testing is generally not indicated.

There is no definitive lower limit of glucose that unequivocally defines pathologic hypoglycemia during a monitored fast. Females tend to have lower fasting glucose levels than males and may have glucose levels as low as 50 mg/dL (2.8 mmol/L) without symptoms.

In patients with a history of a postprandial pattern of hypoglycemia, such as in a patient with hypoglycemia after bariatric surgery, a prolonged fast may not trigger hypoglycemia. In those individuals, hypoglycemia should be assessed after a mixed meal or after eating a meal that is similar to a meal that previously triggered hypoglycemic symptoms (often high in refined carbohydrates and fat). Fingerstick glucose measurements and blood tests for glucose, insulin, and C-peptide can be checked at 30-minute intervals after the meal for up to 4 hours.

Treatment of Hypoglycemia

- Oral sugar or IV dextrose
- Sometimes parenteral glucagon
- Sometimes surgery or other medications, depending on the cause

Treatment in patients taking antihyperglycemic medications

Immediate treatment of hypoglycemia involves provision of glucose. Individuals at risk for hypoglycemia should have glucagon or dasiglucagon at home and elsewhere, and household members and trusted others should be instructed on management of hypoglycemic emergencies. For patients with an underlying hypoglycemic disorder, measures to prevent hypoglycemia will be as important as correcting the blood sugar immediately.

Patients with prolonged or severe hypoglycemia should be hospitalized for continuous intravenous dextrose infusion. In cases of severe hypoglycemia due to sulfonylurea ingestion, octreotide should be administered along with dextrose.

In patients treated with glucose-lowering medications (insulin or sulfonylureas), a plasma glucose level < 70 mg/dL (< 3.9 mmol/L) is considered hypoglycemia and should be treated to avoid further decreases in blood glucose and consequences of hypoglycemia.

Patients who are able to eat or drink can drink juices, sucrose water, or glucose solutions; eat candy or other foods; or chew on glucose tablets when symptoms occur.

The rule of 15s should be followed for treatment of hypoglycemia. Typically, 15 g of glucose or sucrose should be ingested. Patients should check their glucose levels 15 minutes after glucose or sucrose ingestion and ingest an additional 15 g if their glucose level is not > 80 mg/dL (> 4.4 mmol/L). After glucose levels improve to > 80 mg/dL, a snack containing complex carbohydrates and protein may be ingested to prevent the glucose level from dropping again.

Adults and children who are unable to eat or drink can be given glucagon 0.5 mg (< 25 kg body weight) or 1 mg (≥ 25 kg) subcutaneously or intramuscularly. Glucagon nasal spray, 1 actuation (3 mg) once in 1 nostril may be used in adults and in children ≥ 4 years. Dasiglucagon, 0.6 mg once subcutaneously (available in an autoinjector), may also be used in adults and in children ≥ 6 years.

Adults or adolescents who are hospitalized can be treated with 50% dextrose, 50 to 100 mL IV bolus, with or without a continuous infusion of 5 to 10% dextrose solution until symptoms resolve.

The efficacy of glucagon depends on the size of hepatic glycogen stores; glucagon has little effect on plasma glucose in patients who have been fasting or who are hypoglycemic for long periods.

Hyperglycemia may follow hypoglycemia either because too much sugar was ingested or because hypoglycemia caused a surge in counter-regulatory hormones (glucagon, epinephrine, cortisol, growth hormone).

Treatment in patients not taking antihyperglycemic medications

Hypoglycemia in patients not taking insulin or a sulfonylurea should also be corrected with oral sugar, IV dextrose, or glucagon.

Underlying disorders causing hypoglycemia must also be treated. Islet cell and non–islet cell tumors must first be localized, then removed by enucleation or partial pancreatectomy; approximately 6% recur within 10 years. Diazoxide and octreotide or other somatostatin analogs can be used to control symptoms while the patient is awaiting surgery or when a patient refuses or is not a candidate for a procedure.

Non-insulinoma pancreatogenous hypoglycemia syndrome (NIPHS) is most often a diagnosis of exclusion after an islet cell tumor is sought but not identified. NIPHS is more often associated with postprandial hypoglycemia, similar to post-gastric bypass hypoglycemia.

Patients with postprandial hypoglycemia due to NIPHS or following gastric bypass can sometimes be treated with frequent low carbohydrate meals, but other treatments such as acarbose to decrease the rate of postprandial glucose absorption and resulting insulin response may be used. Verapamil, diazoxide, and octreotide have also been used to decrease insulin secretion.

Medications that cause hypoglycemia, including alcohol, must be stopped.

Treatment of hereditary and endocrine disorders, liver failure, renal failure, heart failure, and sepsis are described elsewhere.

Key Points

- In patients taking medications for treatment of diabetes, hypoglycemia is defined as a plasma glucose level < 70 mg/dL (< 3.9 mmol/L).
- Most hypoglycemia is caused by medications used to treat diabetes mellitus (including surreptitious use); insulin-secreting tumors are rare causes.
- To diagnose a hypoglycemic disorder not due to diabetes treatment requires a low plasma glucose level (< 55 mg/dL [< 3 mmol/L]) *plus* simultaneous hypoglycemic symptoms that reverse with dextrose administration.
- If the etiology of hypoglycemia is unclear, do a 48- or 72-hour fast with measurement of plasma glucose at regular intervals and whenever symptoms occur.
- Measure serum insulin, C-peptide, and proinsulin at times of hypoglycemia to distinguish endogenous from exogenous (factitious) hypoglycemia.
- Treat patients with hypoglycemia due to insulin or antihyperglycemic medications with oral or parenteral glucose and glucagon, depending upon clinical severity.
- In patients with hypoglycemia not due to insulin or other antihyperglycemic medications, give glucose and glucagon and treat the underlying disorder.

Drug Information for the Topic